SQLite vs Postgres:
single vs cluster HTTP benchmark

Riset empiris degradasi persentil WAL SQLite pada Go Fiber dan Bun reusePort, serta rekomendasi arsitektur production untuk aplikasi read-heavy single-server.

10 September 2026 · Data & kode: github.com/maulanashalihin/sqlite-cluster-bench

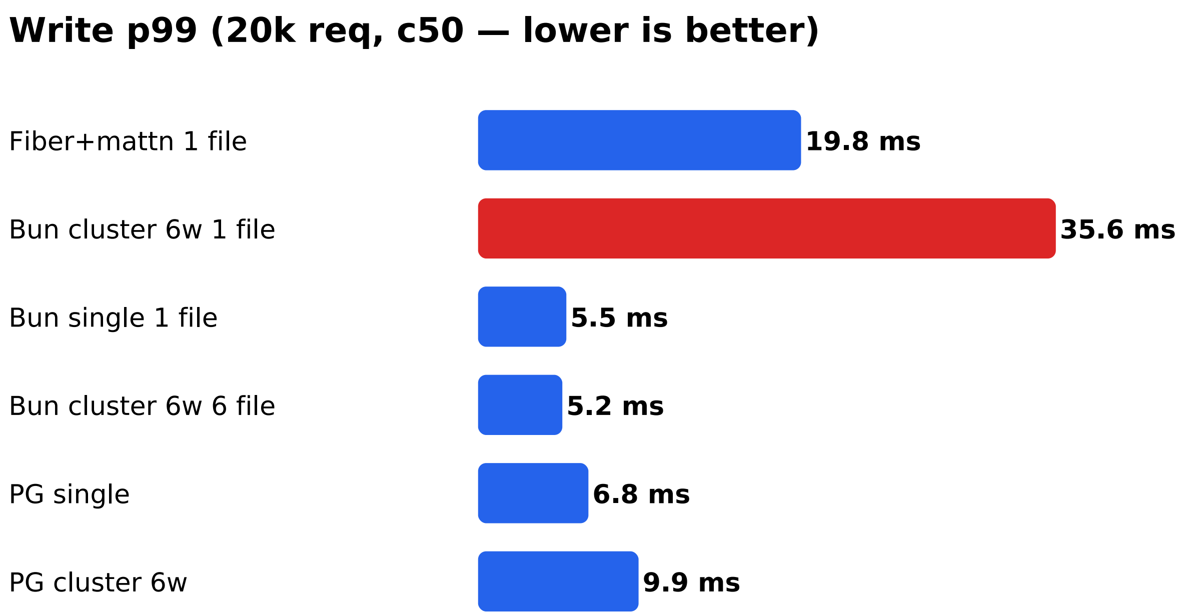
1. Ringkasan eksekutif

1. **Bun cluster mengalami hal serupa — tail write lebih lebar.** Penyebabnya bukan goroutine, melainkan satu WAL write-lock SQLite: 6 proses berebut gembok lintas-proses.
2. **Single event-loop = antrean single-writer gratis** (write p99 5,5ms vs 35,0ms cluster-1-file).
3. **reusePort obat pintu, bukan dapur**: SQLite read +73%, PG c50 −10%.
4. **Sharding (1 worker 1 file)**: write +86%, tail rata.
5. **Postgres tanpa anomali tail** (gap p99 ≤1,5x); crossover cluster di c150.
6. **Rekomendasi: SQLite, 1 proses writer, WAL + synchronous NORMAL + busy_timeout 5000.**

2. Metodologi

Mesin 6 CPU / RAM 11,7 GB Ubuntu · Go 1.27.0, Fiber v2.52.9, mattn/go-sqlite3 v1.14.32 (CGo), pgx v5.11.0 · Bun 1.4.0 · Postgres 18.6 · Skema kv(id PK AUTOINCREMENT, val TEXT 100B), seed 20.000 · SQLite: WAL, synchronous NORMAL, busy_timeout 5000 · Endpoint POST /write (1 INSERT) dan GET /read?id= (point lookup) · Semua run fail=0. Semua angka = mean 3–5 iterasi.

3. E1 — Fiber+mattn vs Bun cluster, write (20k, c50, 5 iterasi)



	rps	p50	p95	p99	max
READ Fiber	46.300	0,92	2,25	4,06	15,1
READ Bun 6w	51.372	0,82	1,68	2,92	8,3
WRITE Fiber	14.635	2,25	9,24	19,84	192,1
WRITE Bun 6w	15.149	1,04	12,60	**35,59**	103,0

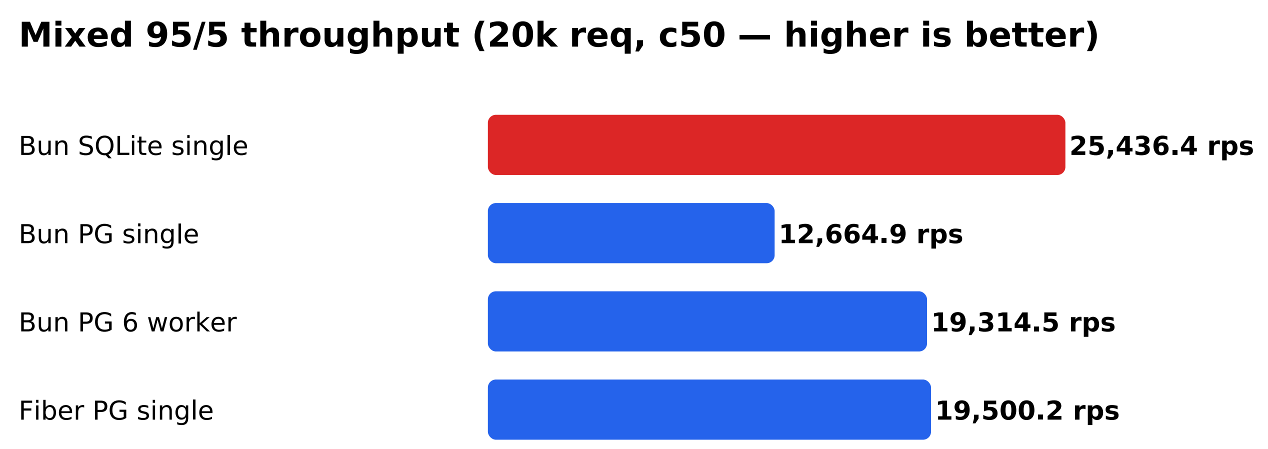
Range p99 write 5 ronde tak overlap (19,2–21,1 vs 34,1–37,9). Median Bun 2x lebih baik, tail 1,8x lebih buruk: bimodal — jalur sepi kilat, jalur tabrakan (lock + retry + checkpoint) dalam.

4. E2 — Bun single vs cluster 1 file · E3 — sharded 6 file

WRITE	rps	p50	p95	p99	max
Single 1 file	20.411	2,23	3,97	5,49	13,6
Cluster 6w 1 file	15.557	0,73	12,67	**34,95**	145,7
Cluster 6w 6 file	39.082	1,07	2,63	5,19	15,4

Single +31% throughput, p99 6x. Shard +86% vs single dengan tail rata — perusak = jumlah writer per file. READ: cluster +73% (47.865 vs 27.568 rps).

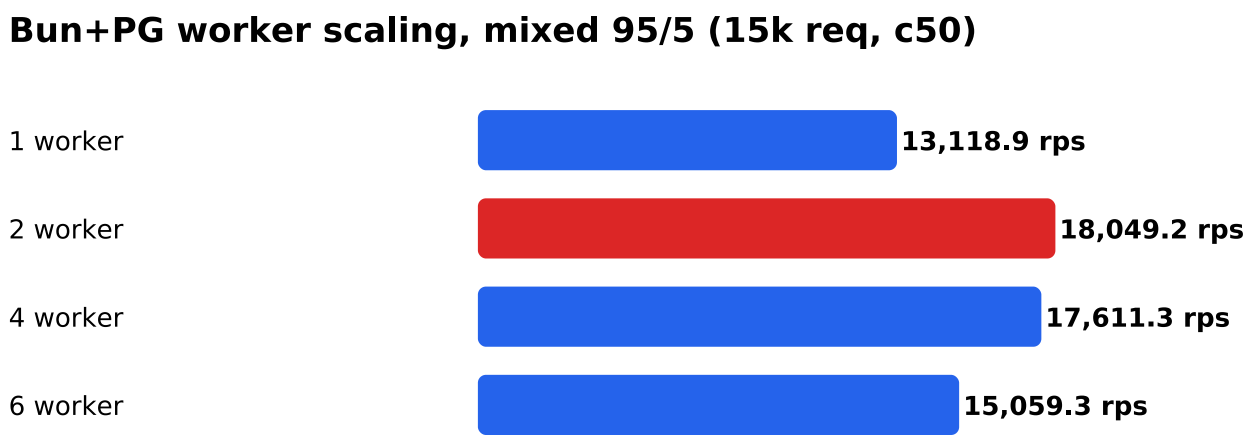
5. E4 — Postgres single vs cluster · E5 — campuran 95/5



Mixed 95/5 (20k, c50)	rps	read p99	write p99
Bun SQLite single	25.436	5,21	5,42
Fiber PG single	19.500	6,98	7,44
Bun PG 6 worker	19.314	10,51	13,39
Bun PG single	12.665	8,00	8,19

PG c50: single sedikit menang (bottleneck ±2ms/query; 60 koneksi cluster jadi overhead). PG c150: cluster +25% read / +17% write — crossover saat accept-queue jenuh. Di backend sama, Fiber single ≈ Bun 6-worker, +55% atas Bun single.

6. E6 — Kurva scaling worker (Bun+PG, mixed 95/5)



U-terbalik: 13.119 → **18.049** → 17.611 → 15.059 rps (1/2/4/6 worker; max 50→210ms). Sweet spot 2 worker.

7. Rekomendasi & guardrail

1. SQLite, tepat 1 proses writer; Bun single atau Fiber+mattn.
2. Write batch dalam transaksi; busy_timeout 5000; monitor ukuran WAL + durasi checkpoint.
3. Sadari tradeoff synchronous NORMAL (jendela kecil kehilangan data saat OS crash).
4. Migrasi ke PG bila write >20–30%, butuh failover, atau multi-node: Fiber single atau Bun 2-worker (bukan 6).

Batas riset: run detik (tanpa WAL-growth jangka panjang); pool 10/worker arbitrer; payload 100B; satu mesin 6-CPU; Fiber+SQLite mixed belum diuji. Reproduksi: lihat README repo.